

Vinh Ly

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Summary

Software Engineer with **18 years** of end-to-end ownership of high-traffic video, e-commerce, and search platforms. Regularly on-call for kernel-level performance tuning of services built on **C++**, **Go**, **Java**, and **Kubernetes**. Delivered a **30 % latency reduction** for a global live-streaming pipeline serving over 200 M daily active users. Partnered with product, data science, and security teams to ship features across YouTube, Google Search, and Microsoft Azure-backed services. Next focus: driving reliable, low-latency integrations for healthcare EHR ecosystems. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages & Scripting: TypeScript, JavaScript, HTML, CSS, SQL, Python

Systems & Networking: AWS

Automation & CI/CD: Terraform, GitHub Actions

Data & Containers: Docker, Postgres, Redis

Work Experience

Staff Software Engineer, YouTube (02/2025 - Present)

- Architected a multi-region live-stream ingest pipeline using **Kubernetes**, **gRPC**, and **Go**, cutting end-to-end latency by 30 % and supporting 200 M daily viewers.
- Led on-call rotation for the core video transcoding service, implementing kernel-level memory optimizations in **C++** that reduced CPU usage by 22 %.
- Collaborated with product and security teams to embed end-to-end encryption, achieving compliance with ISO-27001 for all live-stream assets.
- Mentored a cross-functional squad of 12 engineers, driving adoption of **CI/CD** best practices across the video platform.
- Introduced automated performance regression tests in **Python** and **Grafana**, decreasing release-cycle incidents by 40 %.

Staff Software Engineer, Whatnot (05/2024 - 02/2025)

- Re-engineered the real-time auction engine on **Go** and **Kubernetes**, boosting throughput by 45 % during peak sales events.
- Implemented a low-latency notification service using **gRPC** and **Python**, cutting user-perceived delay from 2 s to 0.6 s.
- Owned on-call responsibilities for the payment processing pipeline, introducing kernel-level socket tuning that reduced error rates by 18 %.
- Partnered with data science to embed real-time fraud detection, decreasing fraudulent transactions by 27 %.
- simplified deployment pipelines with **Docker** and **CI/CD**, cutting release time from days to hours.

Staff Software Engineer/Manager, Google (03/2015 - 04/2024)

- Directed a team of 20 engineers building the core search indexing pipeline on **C++** and **Java**, achieving a 15 % reduction in index latency.
- Designed a distributed caching layer using **Spanner** and **Pub/Sub**, improving query hit rate by 12 % across global data centers.
- Managed on-call rotation for the ad-serving platform, introducing kernel-level thread-pool tuning that cut CPU spikes by 25 %.

- Delivered cross-functional feature releases in partnership with product, UX, and security, maintaining 99.99 % service availability.
- Co-authored internal observability standards, integrating **Prometheus** and **Grafana** dashboards that reduced MTTR by 30 %.

Software Engineer, Google (02/2011 - 02/2015)

- Built high-throughput data ingestion services in **Java** and **Python**, scaling to process 5 TB/day of user activity logs.
- Optimized query execution paths in **C++**, reducing average response time by 18 % for core search features.
- Collaborated with infrastructure teams to migrate services to **Google Cloud Platform**, achieving a 20 % cost reduction.
- Implemented automated testing frameworks with **CI/CD**, decreasing regression bugs by 35 %.
- Provided on-call support for critical search APIs, handling incident triage and root-cause analysis.

Software Design Engineer II, Microsoft (10/2007 - 12/2011)

- Worked closely with partner engineering groups to integrate storage APIs into Office 365, expanding adoption by 40 %.
- Developed core components of the Azure storage stack in **C++**, improving durability metrics by 99.9999 %.
- Created monitoring agents using **Python** and **gRPC**, enabling real-time health checks across 10 k+ VMs.
- Led performance tuning initiatives that cut I/O latency by 22 % for enterprise customers.
- Participated in on-call rotation, handling kernel-level debugging of storage drivers.

Software Design Engineer Intern, Microsoft (06/2006 - 08/2006)

- Implemented a prototype caching layer in **C++** for internal services, demonstrating a 15 % speedup in data retrieval.
- Collaborated with senior engineers to integrate the prototype into the build system using **CI/CD** pipelines.
- Presented findings to the product team, influencing the roadmap for future caching solutions.

Education

B.S. Honors in Computer Science

Virginia Tech | 2004 - 2007